



Karnataka Law Society's Gogte Institute of Technology, Belagavi

(A Permanently Affiliated and Autonomous Institute under Visvesvaraya Technological University, Belagavi | Accredited by NAAC A+ & NBA)



Department of Electrical and Electronics Engineering Project Phase 0

“EV Charging Station with Smart Payment using UPI System ”

Project Type: Industry Based

Name	USN
Sanjana Sajjan	2GI23EE041
Saniya Pinjar	2GI23EE040
Toufiq Nadaf	2GI23EE054
Vinayak Baliger	2GI23EE057

Under the Guidance of:

Prof.Nikhil Chitragar

Assistant Professor Dept. of EEE, KLSGIT, Belagavi

Department of Electrical and Electronics Engineering,
KLS GIT, Belagavi

Content:

Ref. No	Title
1	Problem Statement
2	Introduction
3	Objectives
4	Methodology
5	Literature survey
6	Project schedule week wise
7	Conclusion
8	References

Problem Statement:

To design a smart EV charging system that integrates UPI-based digital payment and enables automatic charging based on payment confirmation, improving efficiency and user convenience

Introduction

- The rapid growth of Electric Vehicles (EVs) has increased the demand for smart and efficient charging infrastructure.
- An EV charging station supplies electrical energy to recharge electric vehicles safely and efficiently
- Integrating a UPI-based smart payment system enables cashless, fast, and secure transactions.
- Users can scan a QR code using UPI apps like PhonePe or Google Pay to make payments.
- After payment confirmation, the system automatically activates the charging process using a microcontroller (ESP32).
- This system supports Digital India, green mobility, and sustainable energy solutions.

Objectives

- To design and develop a smart EV charging station
- To integrate UPI-based digital payment system
- To automate charging after successful payment
- To provide a user-friendly and efficient system
- To reduce dependency on manual supervision

Methodology

- The system displays a unique UPI QR code for payment.
- The user scans the QR code using a UPI application and completes the payment.
- Payment confirmation is received through API/SMS/Cloud server verification.
- Once payment is verified, the ESP32 microcontroller activates the relay module and the charging process continues based on the amount paid.
- The relay connects the power supply to start EV charging.
- Voltage and current sensors continuously monitor charging parameters.
- After the allocated charging time or energy limit is reached, the system automatically stops charging.

BLOCK DIAGRAM

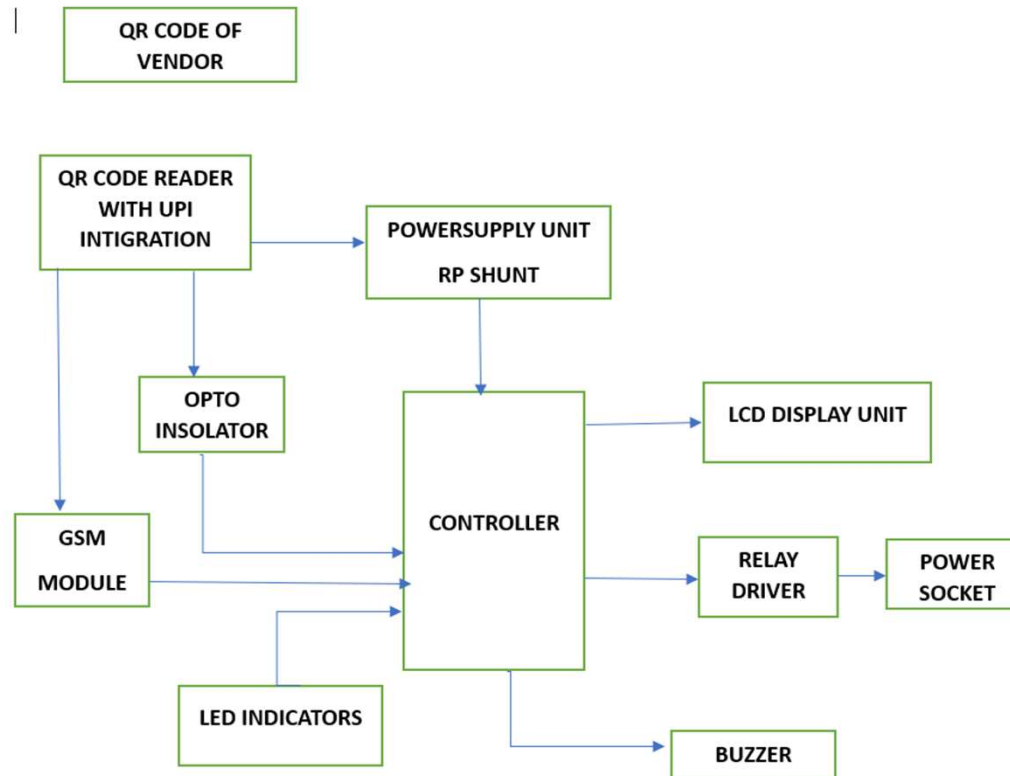


Fig 1.1

Department of Electrical and Electronics Engineering,
KLSGIT, Belagavi

Literature Survey

Ref. No	Title of the Paper	Published Year	Journal / Conf Name	About the paper	Gaps identified
[1]	Smart EV Charging Station with Automated Payment System	2025	International Journal of Scientific Research in Engineering and Technology (IJSRSET)	Proposes an IoT-based EV charging system with automated billing and monitoring. Focuses on improving charging efficiency and reducing manual intervention.	Does not include UPI-based payment; relies on traditional digital payment methods
[2]	Energy Ease:UPI-Based EV Charging Station	2026	Premier Science Journal (PJS)	Presents a smart EV charging station using ESP32 and UPI QR code payment. Enables real-time monitoring and seamless payment experience.	Highly dependent on internet connectivity; lacks offline payment option.
[3]	Electric Vehicle Charging Station with UPI Payment Service	2025	International Journal of Innovative Research (IJIR)	Describes integration of UPI payment in EV charging stations using QR scanning for fast and secure transactions.	Limited discussion on security and scalability of the system.
[4]	Iot-Based Smart EV Charging Station	2024	International Journal of Scientific Research and Technology (IJSRST)	Focuses on IoT-enabled charging with real-time data monitoring, energy optimization, and remote access.	Requires dedicated mobile apps; no integration of UPI or simple payment system.

Project Schedule week wise

Sl no	Type of Outcome	Completed? (Yes/No)
1	Literature survey and Planning	Week 1-2
2	Hardware Procurement and Setup	Week 3-4
3	Payment Integration	Week 5-6
4	Testing and Debugging	Week 7-8
5	Final Documentation	Week 9

Conclusion

- The proposed system provides a smart and automated EV charging solution integrated with UPI-based digital payment.
- It enables secure, fast, and cashless transactions using QR code scanning.
- Automatic activation of charging after payment confirmation improves efficiency and user convenience.
- Monitoring of voltage and current ensures safe and reliable operation.

References

- IEEE Xplore Digital Library – Research papers on Smart EV Charging Systems.
- Springer Journal – IoT Based Electric Vehicle Charging Infrastructure.
- Elsevier Journal – Integration of Renewable Energy with EV Charging.
- NPCI Documentation – Unified Payments Interface (UPI) Technical Guidelines.
- Government of India EV Policy Reports and Digital India Initiative Documents.